



VLA Models II

Course AIAA 4220

Week 12- Lecture 24

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VLA in Real-World Robotics

Challenges real robots face that simulations cannot fully capture:

- **Visual variability**

- lighting changes, shadows
- clutter, occlusion
- camera distortions

- **Robot embodiment differences**

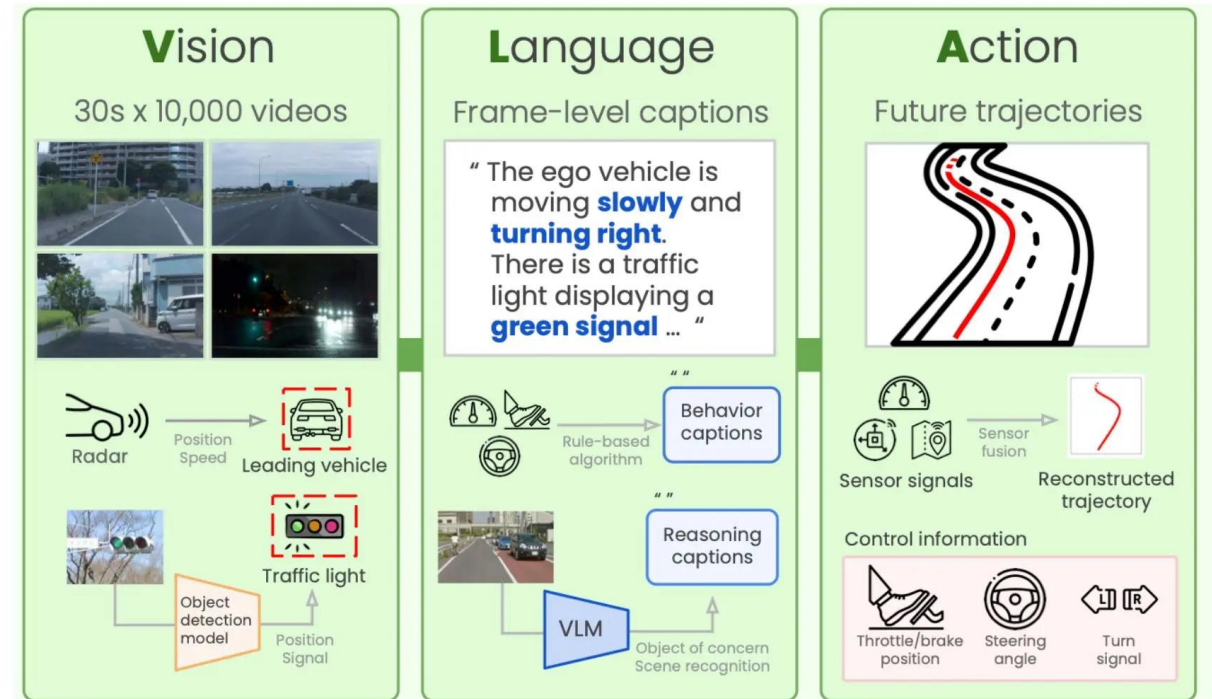
- slightly different kinematics
- wear & tear of motors

- **Sensor noise**

- depth instability
- calibration drift

- **Physical dynamics**

- slippery surfaces
- deformable objects



VLA models improve robustness because they learn **semantic grounding + commonsense priors**.

Representative VLA Models

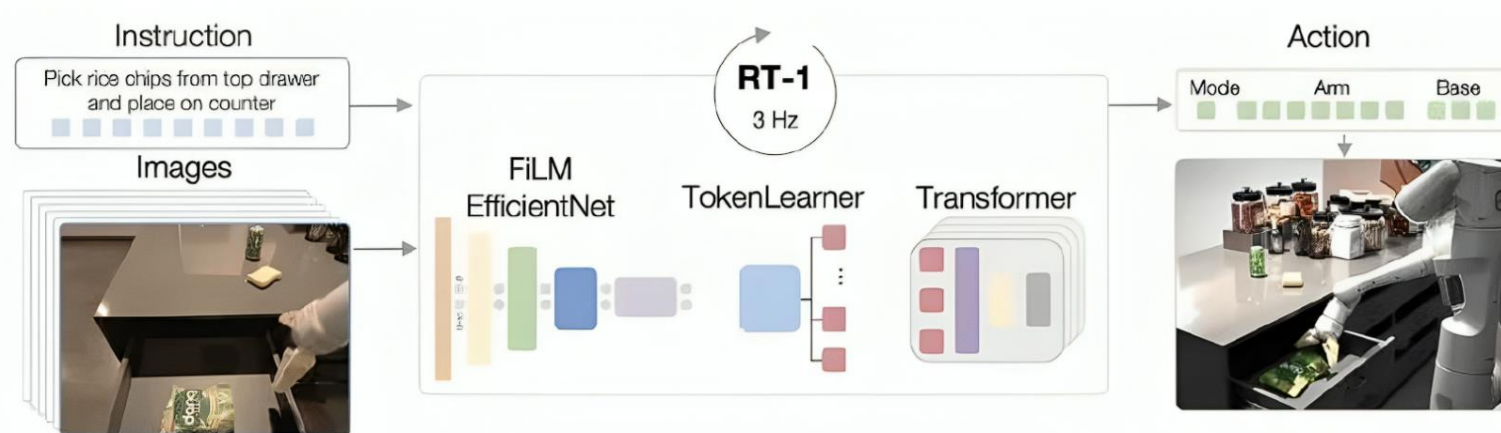
Google: RT-1

- Efficient transformer for robot actions
- 130K demonstrations on household manipulation tasks

Key features:

- Tokenized action representation**
- Vision tokens from MobileNet**
- Fast inference (<20ms)**

Improved real-world reliability.



Traditional robots require extensive human programming or detailed instructions for every task. RT-1, however, leverages a remarkable machine learning concept known as Transformers, initially crafted to understand language by assessing context and relationships within data.

RT-1 extends this capability into physical robotics, enabling robots to process visual and language instructions simultaneously and react quickly in real-time — matching human response speeds at approximately three decisions per second.

Representative VLA Models

Google: RT-2

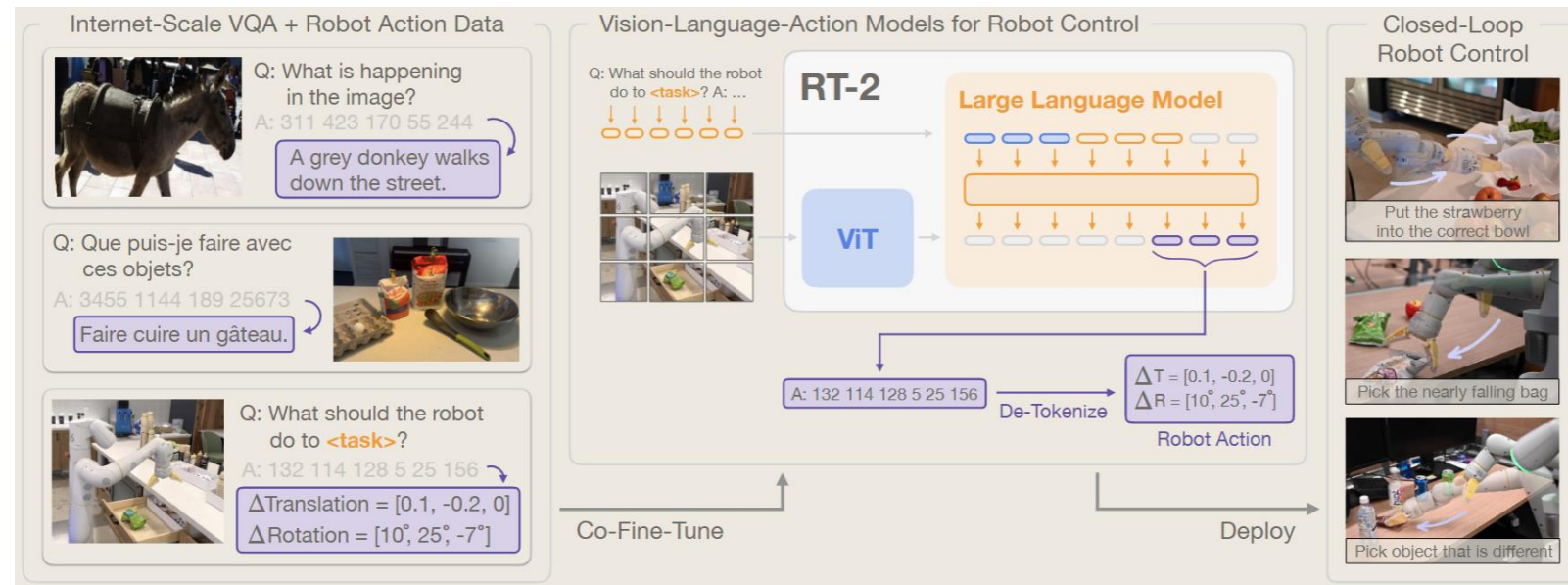
Breakthrough:

- Integrates **robot data + web-scale vision-language data**
- Uses PaLM-E style LLM priors
- Exhibits reasoning capabilities:
 - object affordances
 - semantic relations
 - novel-object generalization

Example task:

- “Throw away the trash.”

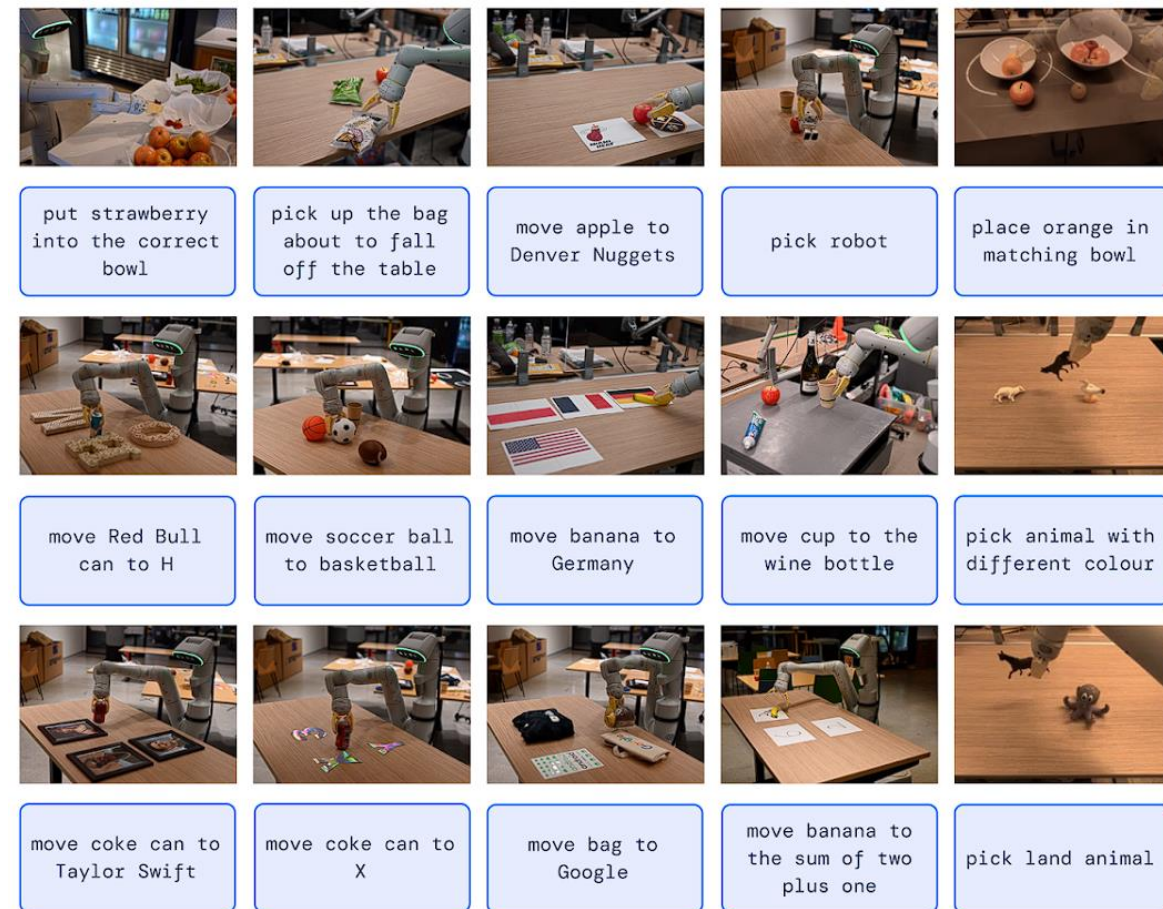
- RT-2 identifies trash → moves in correct direction → manipulates to dispose.



Representative VLA Models

Google: RT-2

- By combining the VLMs' broad knowledge with RT-1's robotics experience, RT-2 gained stronger general skills for understanding and manipulating objects.
- When tested, it could follow commands about novel items better than previous approaches.
- As the researchers put it, VLM pre-training allowed for “considerable benefit” in transferring web knowledge to the robotics domain.



Representative VLA Models

PKU:Uni-NaVid

Uni-NaVid is a versatile multi-task navigation agent, incorporating four widely demanded navigation tasks: vision-and-language navigation, object-goal navigation, embodied question answering, and human following. These tasks are distinct from each other, with varying task settings and objectives.



Representative VLA Models

PKU:Uni-NaVid

Uni-NaVid adopts a foresight prediction to generate actions for a future horizon at once instead of step-by-step. This enables Uni-NaVid to achieve 5Hz inference, facilitating the deployment of a non-blocking navigation robot powered by a VLA model in real-world environments

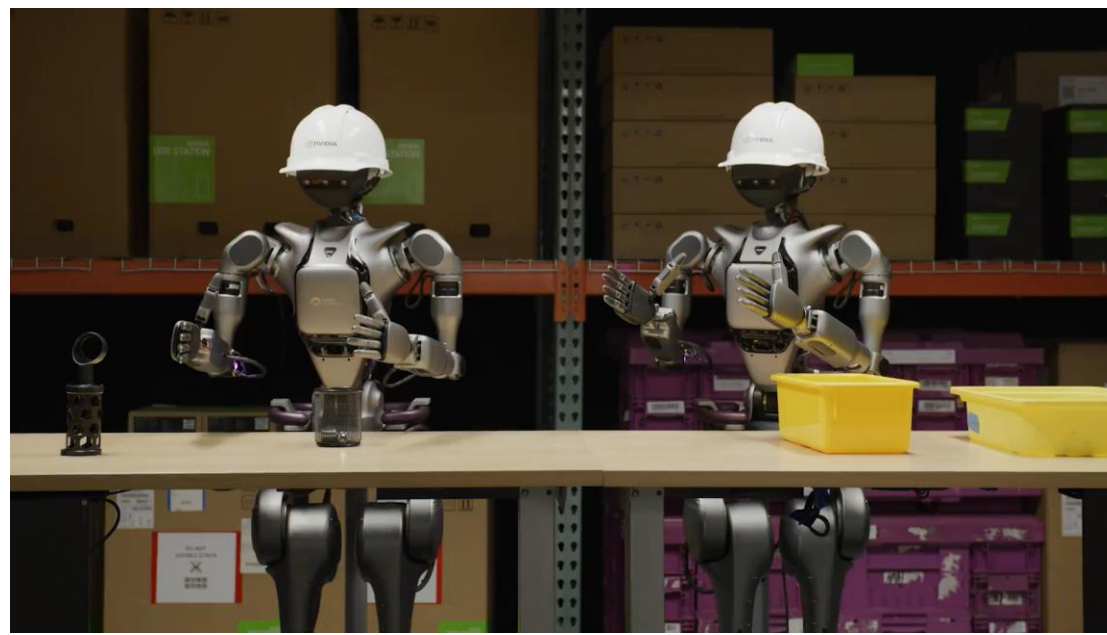
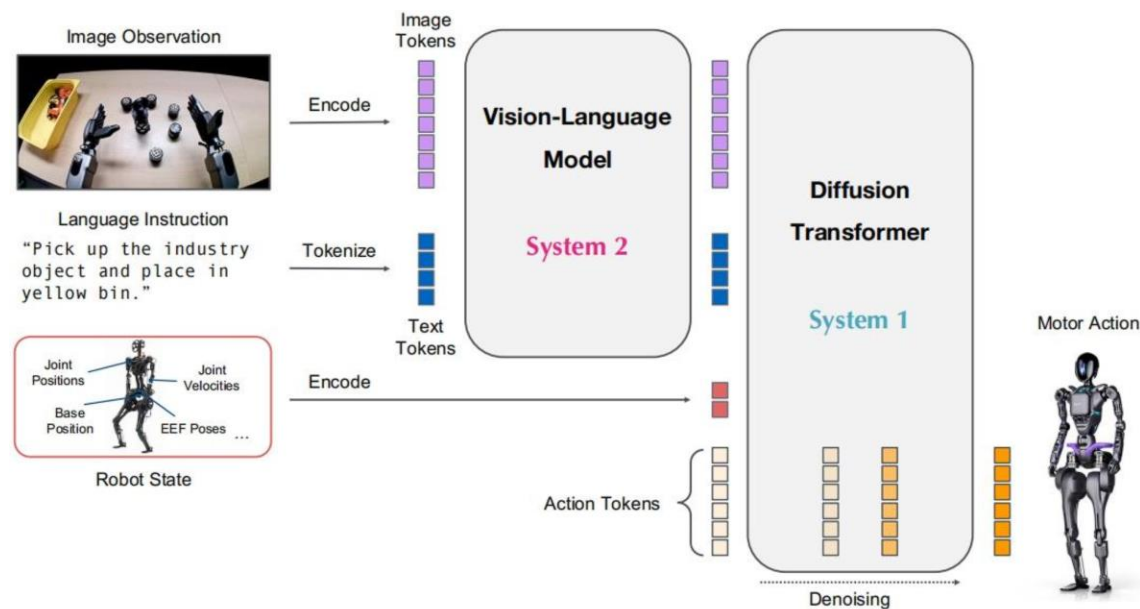
Move to the man on the right side. Then follow the man until you see a sofa. Then turn right search for a TV, stop by the TV.



Representative VLA Models

NVIDIA: Groot N1

- Inspired by dual-process theories of human cognition, models like NVIDIA's Groot N1 implement two complementary subsystems: a fast reactive module (System 1) and a slow-reasoning planner (System 2).
- **System 1 comprises a diffusion-based control policy** that operates at 10 ms latency, ideal for fine-grained, low-level control such as end-effector stabilization or adaptive grasping.
- **System 2 uses a LLM for task planning, skill composition, and high-level sequencing.** The planner parses long horizon goals (e.g., "clean the table") into atomic subtasks, while the low-level controller ensures real-time execution.



Representative VLA Models

Google: SayCan

- SayCan uses pretrained behaviors to ground language models, enabling them to propose feasible natural language actions.
- The robot acts as the language model's "hands and eyes," while the language model supplies high-level task knowledge.
- Low-level tasks are combined with large language models, where value functions provide the grounding necessary to connect high-level knowledge to physical environments.



Scaling Laws for Embodied Intelligence

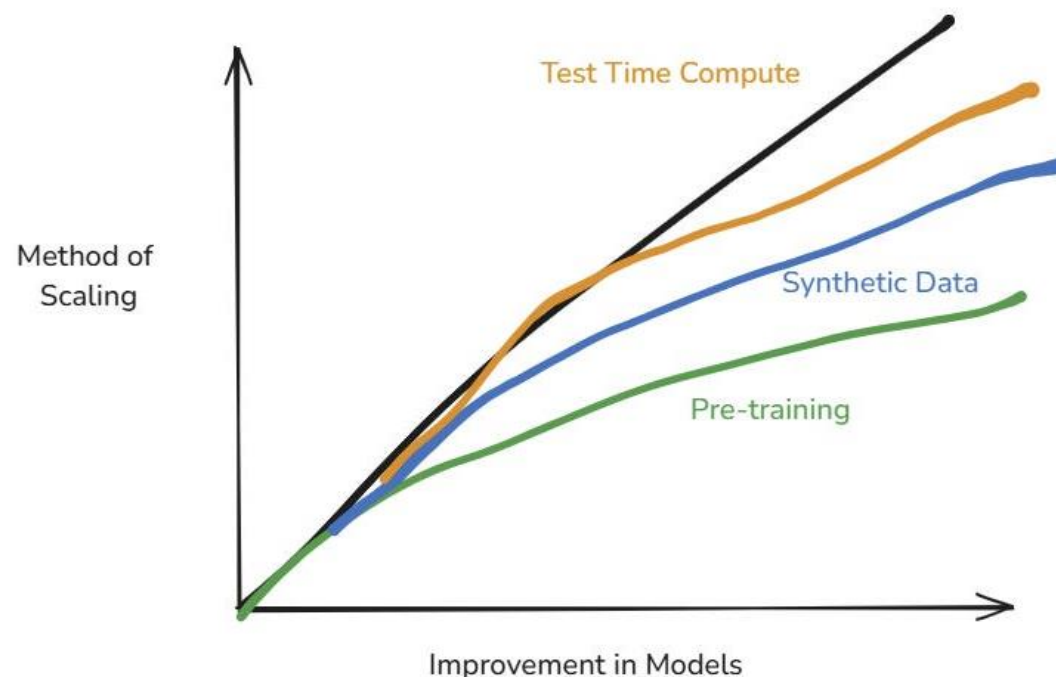
Empirical observations:

- ✓ Increasing **model size** → improves generalization
- ✓ Increasing **demo data** → increases manipulation success
- ✓ Adding **web-scale image-language data** → huge leap in semantic reasoning

Joint scaling of:

- robot trajectories
- multimodal tokens (vision + language)
- model parameters
- training compute

All major VLAs (RT-2, PaLM-E, OpenVLA) follow scaling laws.



Types of Data Needed by VLA Models

1. Robot Demonstration Data

- Human teleoperation
- Kinesthetic teaching
- Play data
- Multi-arm datasets

2. Vision–Language Data

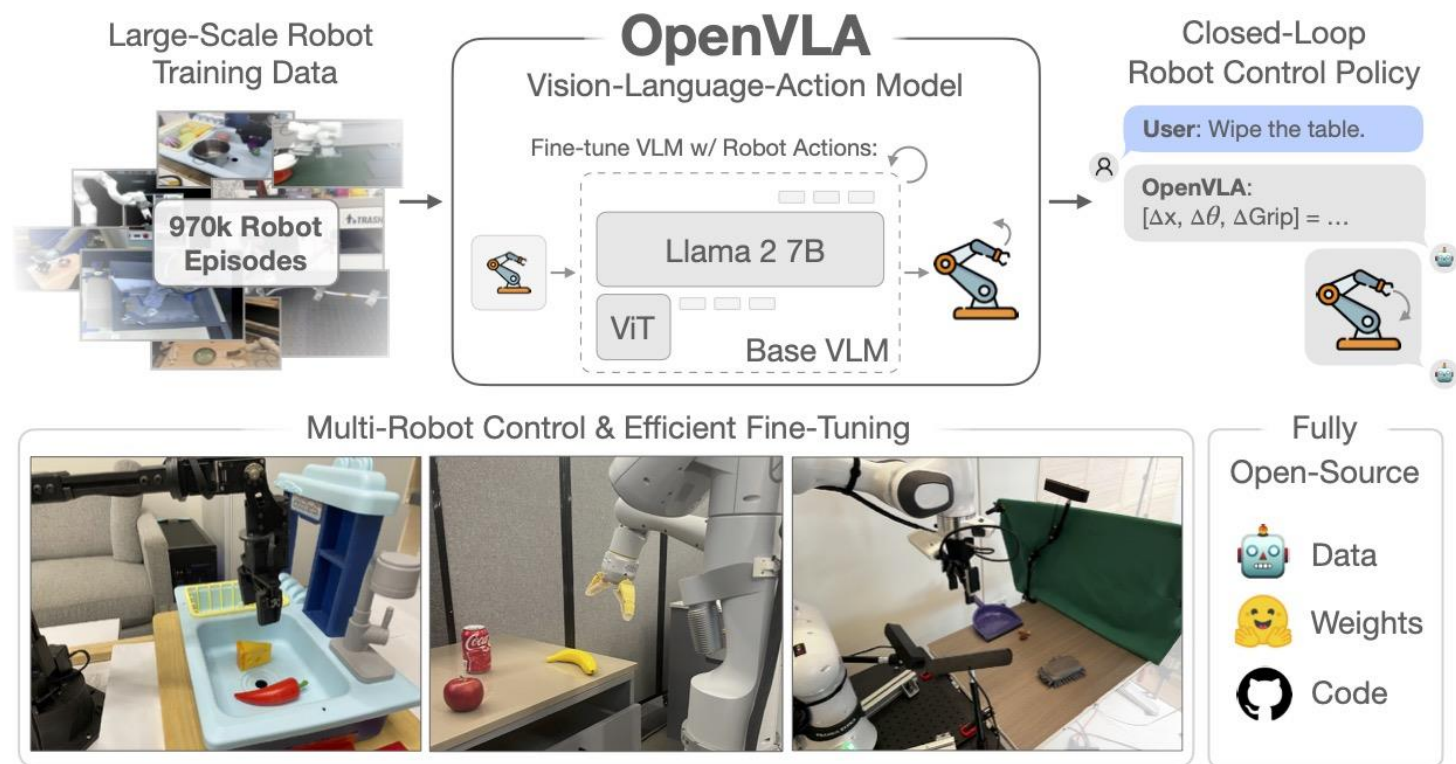
- CLIP-style internet image–text
- Object relations, affordances
- Scene semantics

3. Egocentric / First-Person Data

- Ego4D, EPIC-Kitchens
- Natural human behaviors for imitation

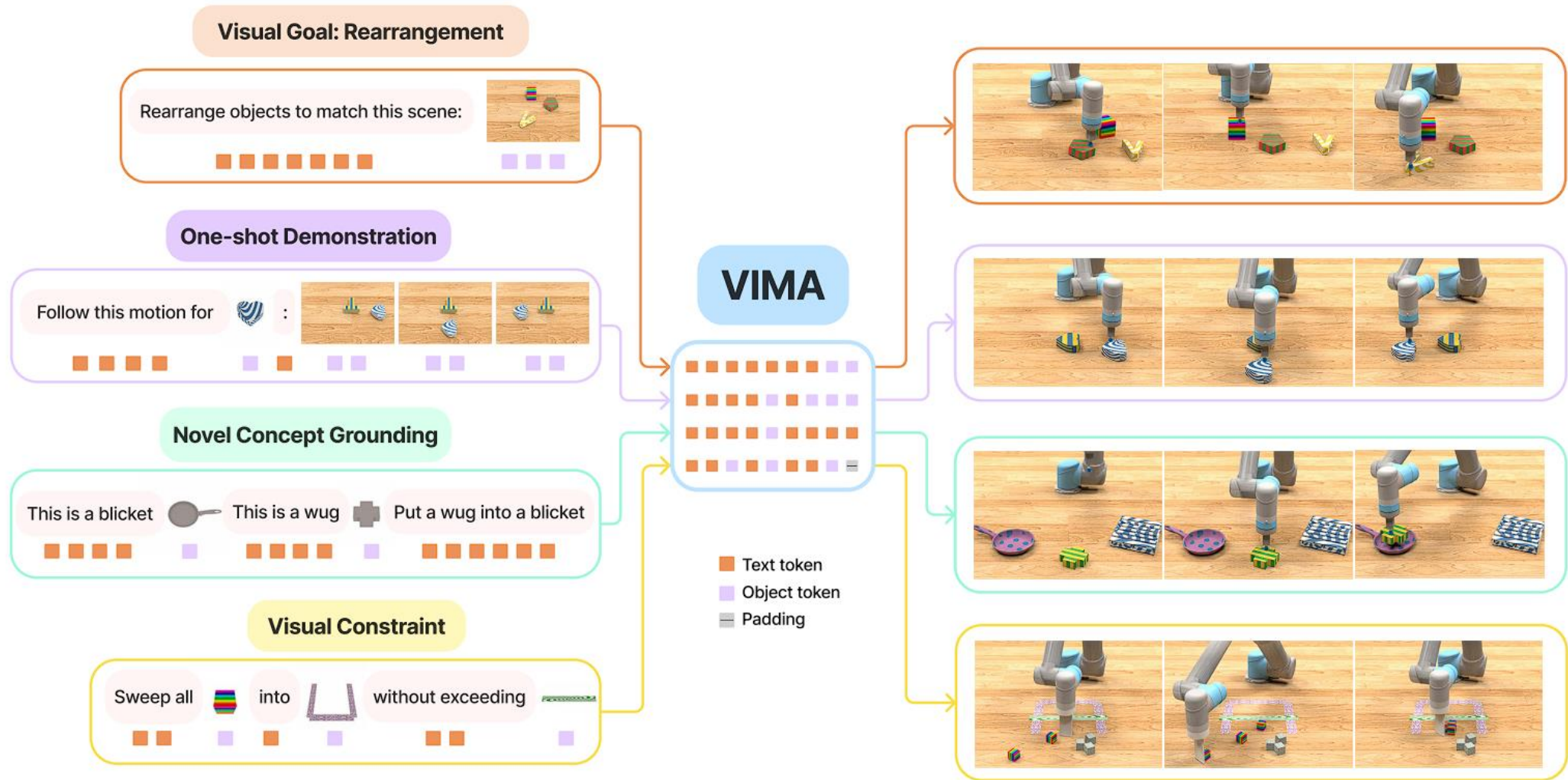
4. Simulation Data

- Isaac Gym
- Habitat
- CALVIN



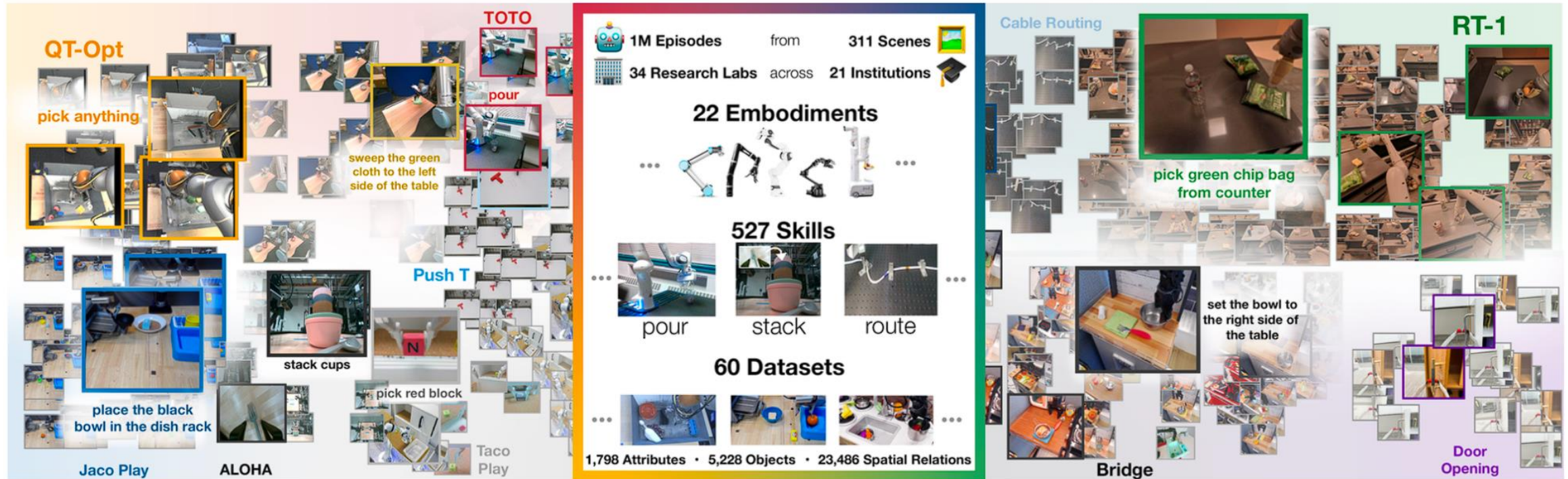
Benchmarks and Simulators

VIMA-Bench



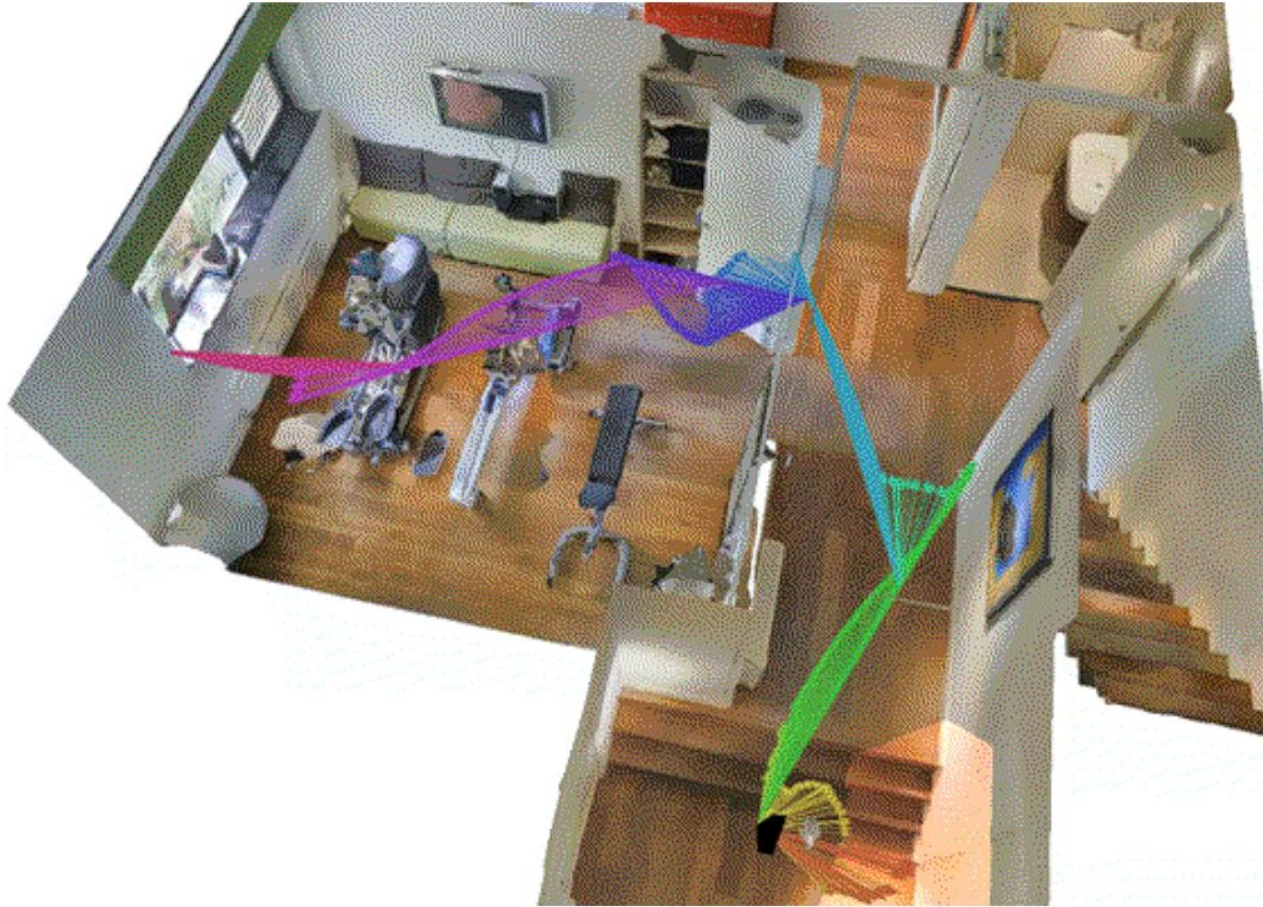
Benchmarks and Simulators

Open X-Embodiment Dataset



Benchmarks and Simulators

Room-To-Room (R2R) & Room-Across-Room (RxR) Dataset

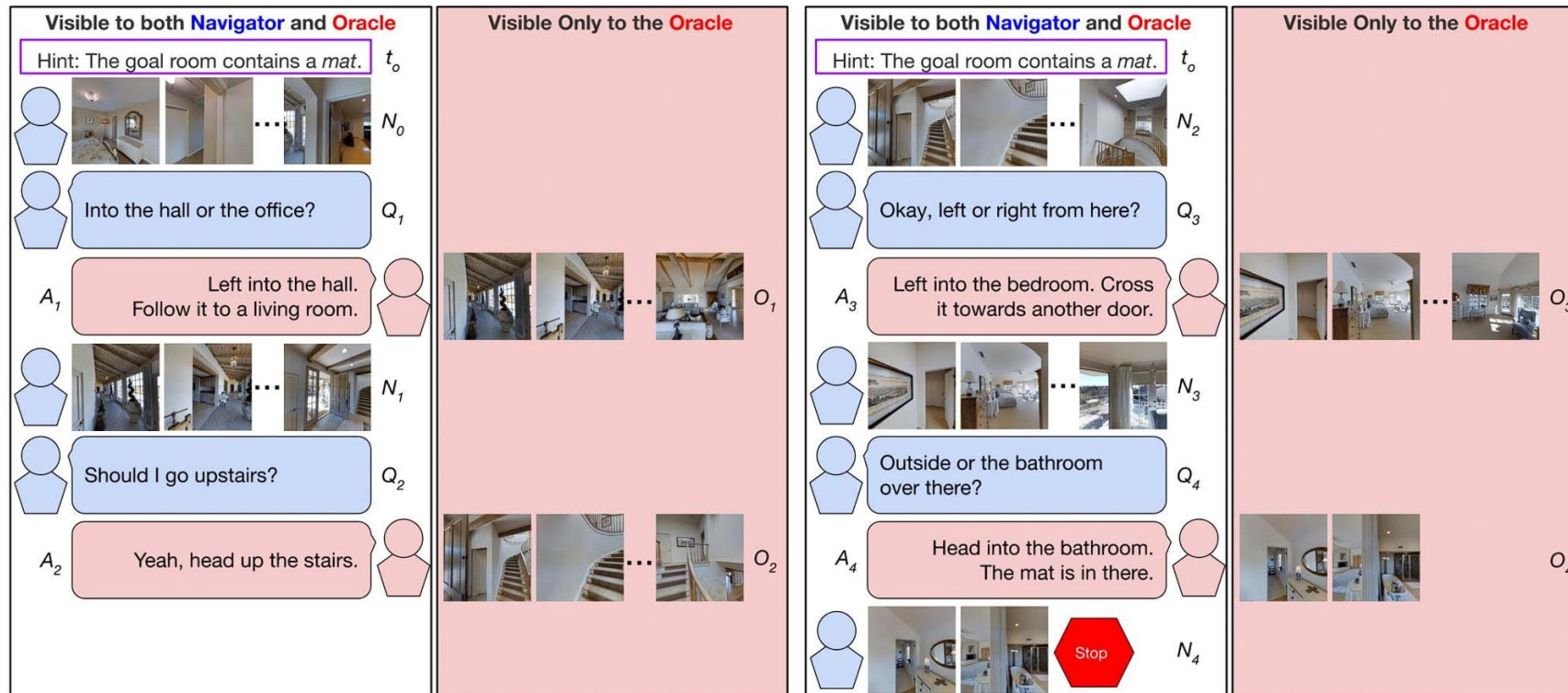


Now you are standing in front of a closed door, turn to your left, you can see two wooden steps, climb the steps and walk forward by crossing a wall paint which is to your right side, you can see open door enter into it. This is a gym room, move forward, walk till the end of the room, you can see a grey colored ball to the corner of the room, stand there, that's your end point.

Benchmarks and Simulators

Cooperative Vision-and-Dialog Navigation (CVDN)

CVDN is a dataset of embodied, human-human dialogs situated in a simulated, photorealistic home environment. The Navigator asks questions of their partner, the Oracle, who has privileged access to the best next steps the Navigator should take according to a full-state information shortest path planner. The dataset consists of 2050 human-human navigation dialogs, comprising over 7k navigation trajectories punctuated by question-answer exchanges across 83 house scans.



Benchmarks and Simulators

Sim-to-Real Transfer

Core challenge: models trained in simulation fail in reality.

Problems:

- Visual gap: sim images too clean
- Physics gap: real-world friction + compliance
- Policy gap: real robots have mechanical imperfections

Solutions:

- domain randomization
- photorealistic rendering
- real-world fine-tuning
- sensory augmentation (depth, tactile)
- representation learning shared between sim & real

- Habitat
- AI2-THOR
- ManiSkill
- Issac Lab



Representation Alignment Problems

VLA models must align:

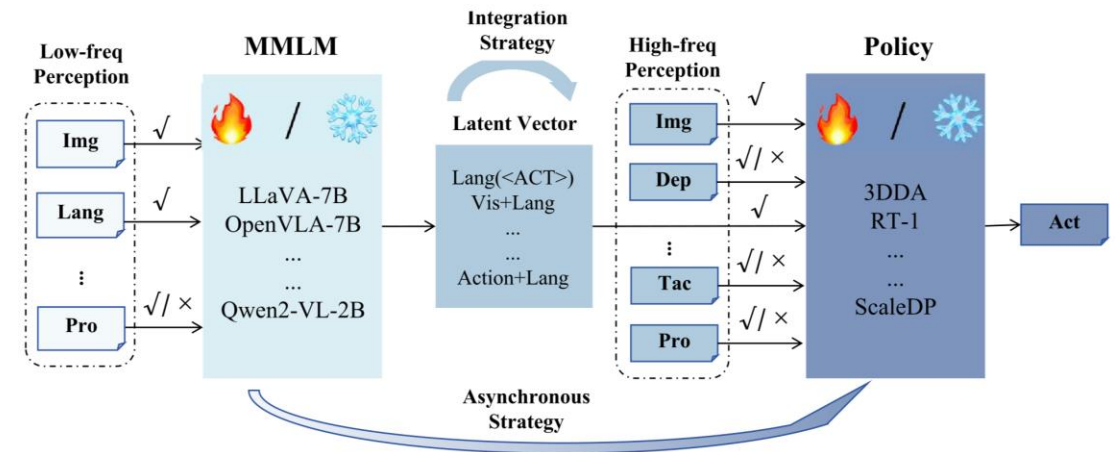
- **Vision tokens** (pixels → objects → meanings)
- **Language tokens** (instructions → semantics → goals)
- **Action tokens** (motor control → skills)

Misalignment causes:

- wrong object selection
- incorrect spatial reasoning
- “semantic drift” (e.g., confusing ‘bottle’ vs ‘cup’)
- hallucinated actions

Often solved via:

- contrastive learning
- cross-modal attention
- grounding losses



Hallucination and Reliability in Embodied LLMs

Common failure modes:

- Producing **unsafe** actions confidently
- Acting on objects not actually present
- Making up nonexistent affordances
- Incorrectly interpreting ambiguous instructions

Why it happens:

- LLM priors overpower visual observations
- Embodied grounding is weak
- Model overfits to training distribution

Mitigation strategies:

- Uncertainty estimation
- Action filtering
- Vision consistency checks
- Safety constraints embedded in policy

Interpretability Techniques

Why interpretability matters:

- Trusting robot decisions
- Debugging model failures
- Ensuring safety

Tools used:

- Attention maps**
 - where language attends in the visual field
- Gradient-based saliency**
 - sensitive regions in images
- Token-level attribution**
 - which words or tokens influence actions
- Object-level grounding visualizations**

Goal: understand how the robot connected instruction → perception → action.

Safety in VLA-Controlled Robots

Challenges:

- Robots can cause harm from small mistakes
- Language ambiguity leads to unsafe behaviors
- VLA models may not understand physical danger

Safety mechanisms:

- Hard motion constraints
- Fail-safe states (robot freezes on uncertainty)
- Human-in-the-loop overrides
- Geofencing and restricted action sets
- Uncertainty-aware planning
- Refusal to act when instruction conflicts with safety rules

Future direction: **Safety-aligned VLA models.**

Prompting Strategies for VLA Agents

VLA models respond strongly to how tasks are phrased.

Examples of effective prompts:

- “Pick up the red cup **carefully**.”
- “Move slowly to the object.”
- “Avoid the fragile vase.”
- “First locate the spoon, then pick it up.”

Prompt types:

- **Descriptive prompts** (describe goals)
- **Constraint prompts** (“without touching others”)
- **Step-by-step reasoning** (“first... then...”)
- **Negative prompts** (“do not push anything over”)

Well-designed prompts → safer, more robust execution.

In-Context Learning for Robots

VLA models can learn new tasks without retraining, by providing examples inside the prompt.

Methods:

- **Text-only demonstrations**
 - “To do task A, follow these steps...”
- **Multimodal examples**
 - image + description pairs
- **Video demonstrations described in text**
- **Object-level instructions**
 - “Here is what grasping a bottle looks like...”

Enables:

- rapid adaptation
- user-friendly customization
- personalized robot behaviors

Hierarchical Reasoning & Long-Horizon Tasks

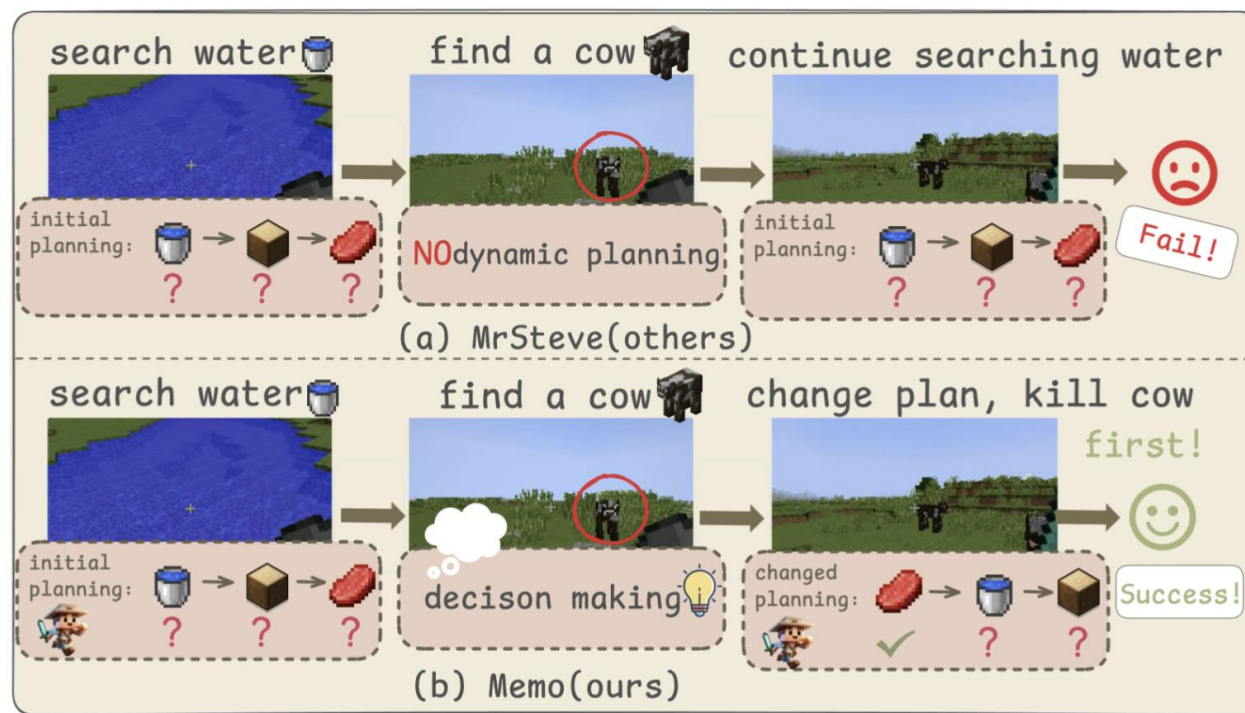
Long-horizon tasks require planning and memory.

Challenges:

- Partial observability
- Multi-step dependencies
- Need for persistent memory
- Error accumulation over long action sequences

Solutions:

- **Hierarchical VLA models**
 - high-level planner (LLM)
 - low-level controller (policy transformer)
- **Skill libraries**
- **Temporal abstractions**
- **LLM chain-of-thought planning**



Multimodal Sensors Beyond Vision

Future VLAs incorporate more modalities:

- **Tactile feedback**

- grasp force
- slip detection

- **Audio cues**

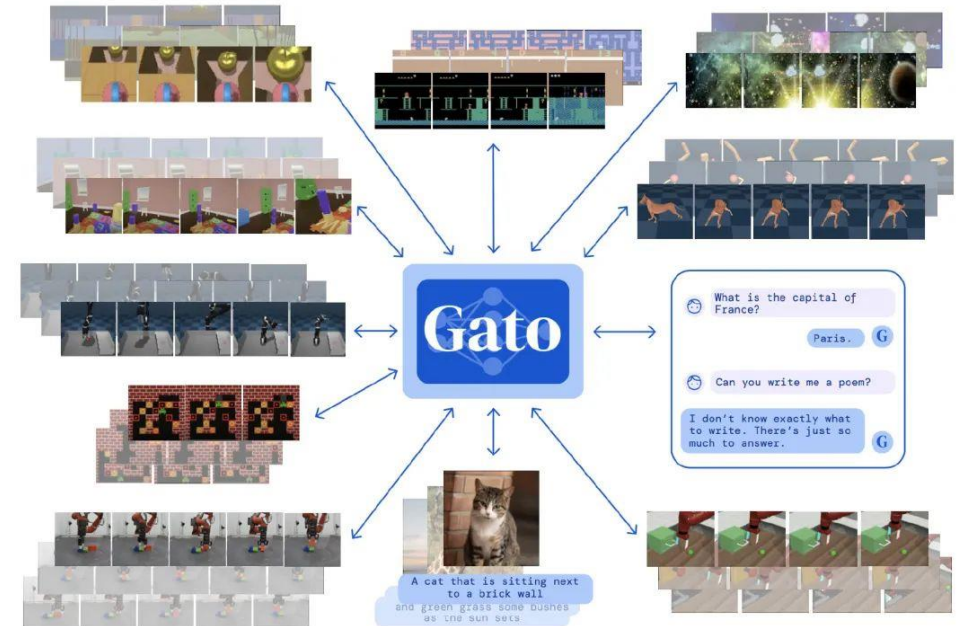
- impact detection
- tool contact

- **Inertial sensors**

- for locomotion and navigation

- **Depth perception**

Goal: robust multimodal grounding for manipulation + navigation.



Practical Deployment Challenges

Issues when running VLA models on real robots:

- 1) Inference latency
- 2) Hardware constraints (edge devices)
- 3) Energy consumption
- 4) Real-time control stability
- 5) Reliability and repeatability
- 6) Calibration drift
- 7) Network connectivity

Solutions:

- model compression
- distillation of large VLA → small real-time VLA
- hybrid classical + learned controllers

Reading Material

- [1] BROHAN A, BROWN N, CARBAJAL J, et al. RT-1: Robotics Transformer for Real-World Control at Scale[J]. 2022.
- [2] BROHAN A, BROWN N, CHEBOTAR Y, et al. RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control[J].
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- [5] ZHANG J, WANG K, WANG S, et al. Uni-NaVid: A Video-based Vision-Language-Action Model for Unifying Embodied Navigation Tasks[J]. 2025.
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Thanks for your attention!

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